Supplementary Material

Understanding the supramolecular chemistry of lutein with gelatin, octenyl succinic anhydride -modified starch polymers in simulated gastro-intestinal buffers: A molecular dynamics study

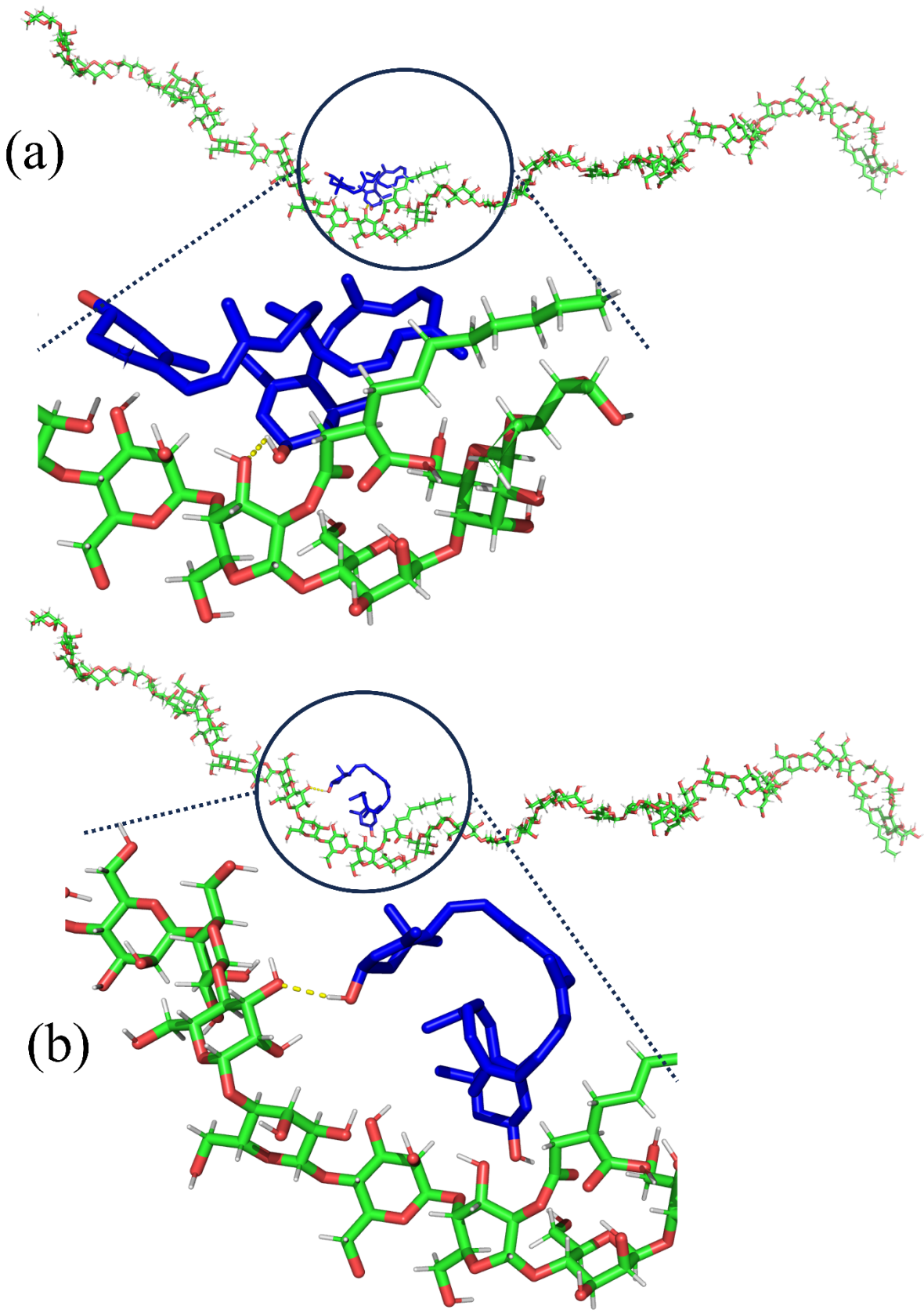
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Table S1. System details

System name	r_{shell} (nm)	*CN*		
		PBS	AIF	AGF
RDF_SOL_HW_ALA_N	0.44	0.032	0.030	0.032
RDF_SOL_HW_ALA_O	0.30	0.004	0.004	0.004
RDF_SOL_HW_ARG_O	0.30	0.004	0.004	0.004
RDF_SOL_HW_CL	0.30	0.010	0.010	0.022
RDF_SOL_HW_GLU_O	0.30	0.004	0.004	0.004
RDF_SOL_HW_GLU_OE1_OE2	0.30	0.020	0.018	0.019
RDF_SOL_HW_GLY_O	0.30	0.013	0.013	0.014
RDF_SOL_HW_HYP_O	0.30	0.004	0.004	0.005
RDF_SOL_HW_HYP_OD1	0.30	0.004	0.004	0.004
RDF_SOL_HW_LYS_NZ	0.44	0.031	0.031	0.032
RDF_SOL_HW_LYS_OC1_OC2	0.30	0.022	0.022	0.020
RDF_SOL_HW_PRO_O	0.30	0.010	0.009	0.010


(a)
(b)

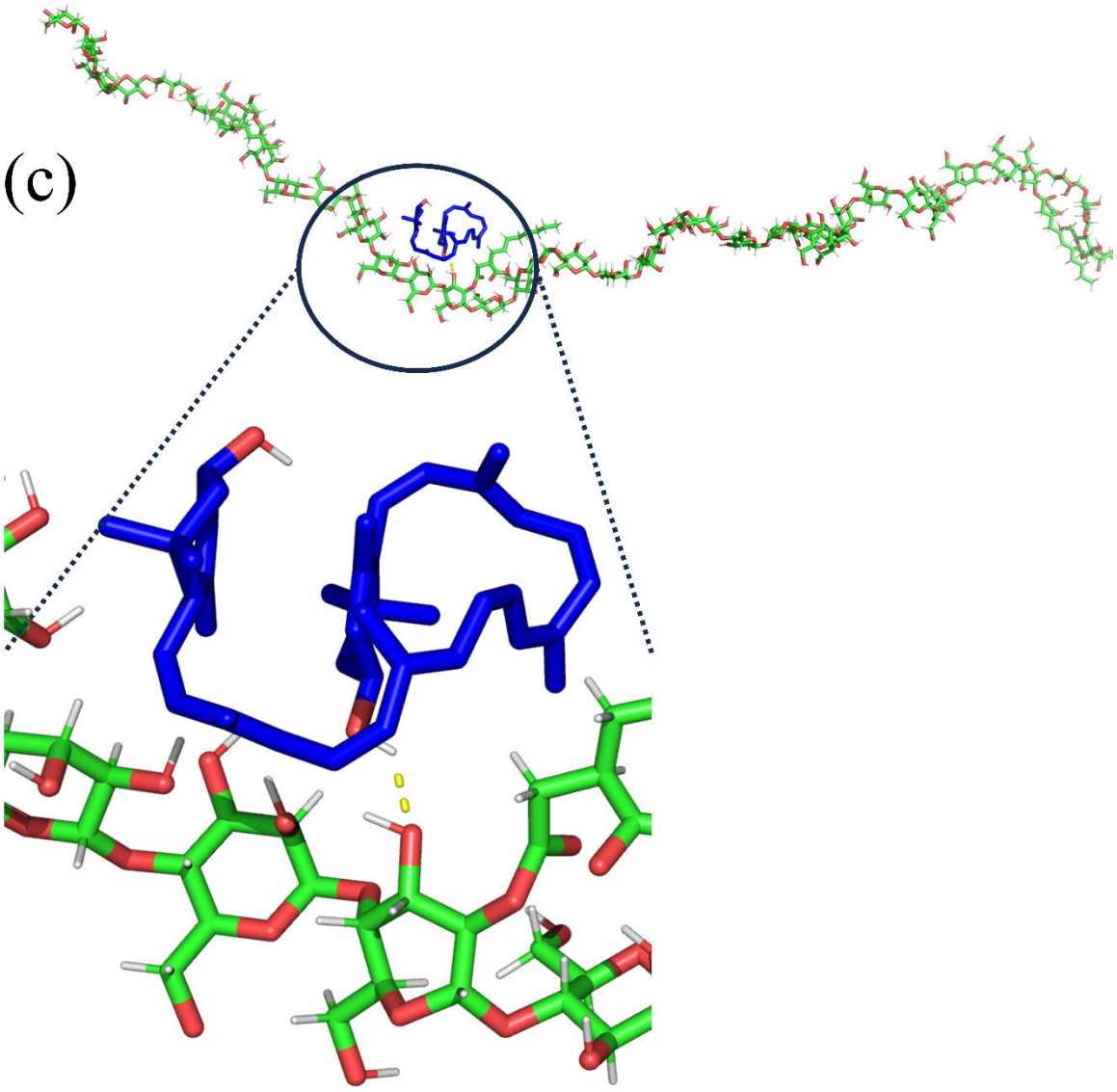


Figure S1. Molecular docking of lutein in OSA-modified starch in different docking positions (a) Docking score at -3.9 (b) docking score at -3.8 with lower rmsd (c) docking score at -3,8 with higher rmsd

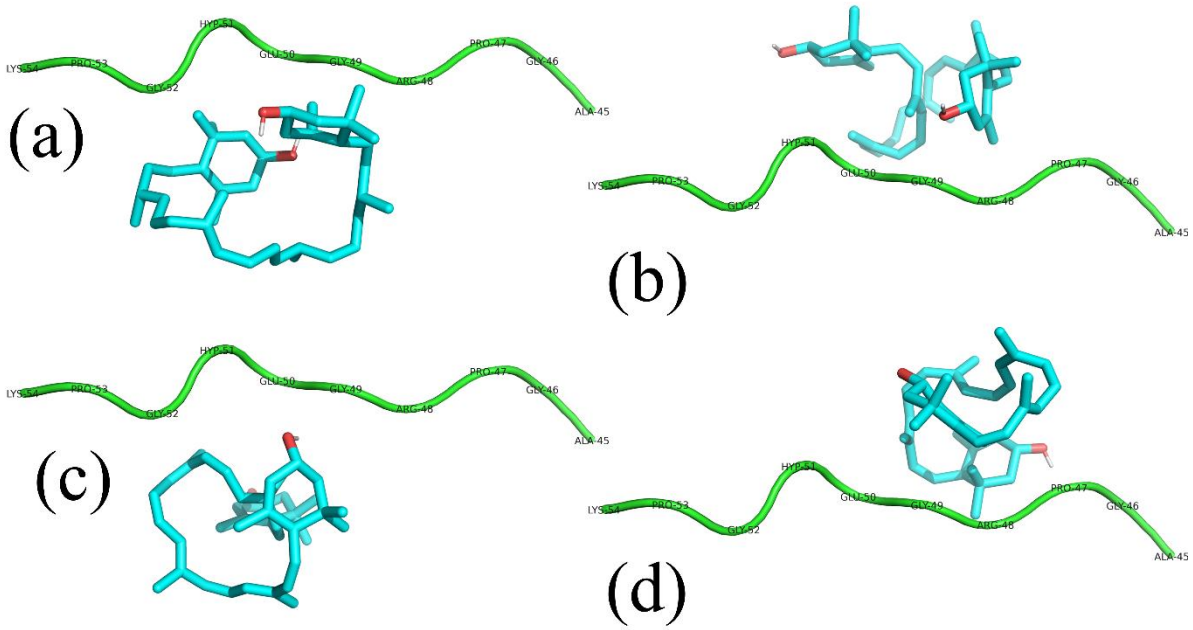


Figure S2. Molecular docking of lutein in gelatin in different docking positions (a) docking score at -3.2 (b) docking score at -3.0 (c) docking score at -2.9 with a lower rmsd (d) docking score 2.9 at higher rmsd

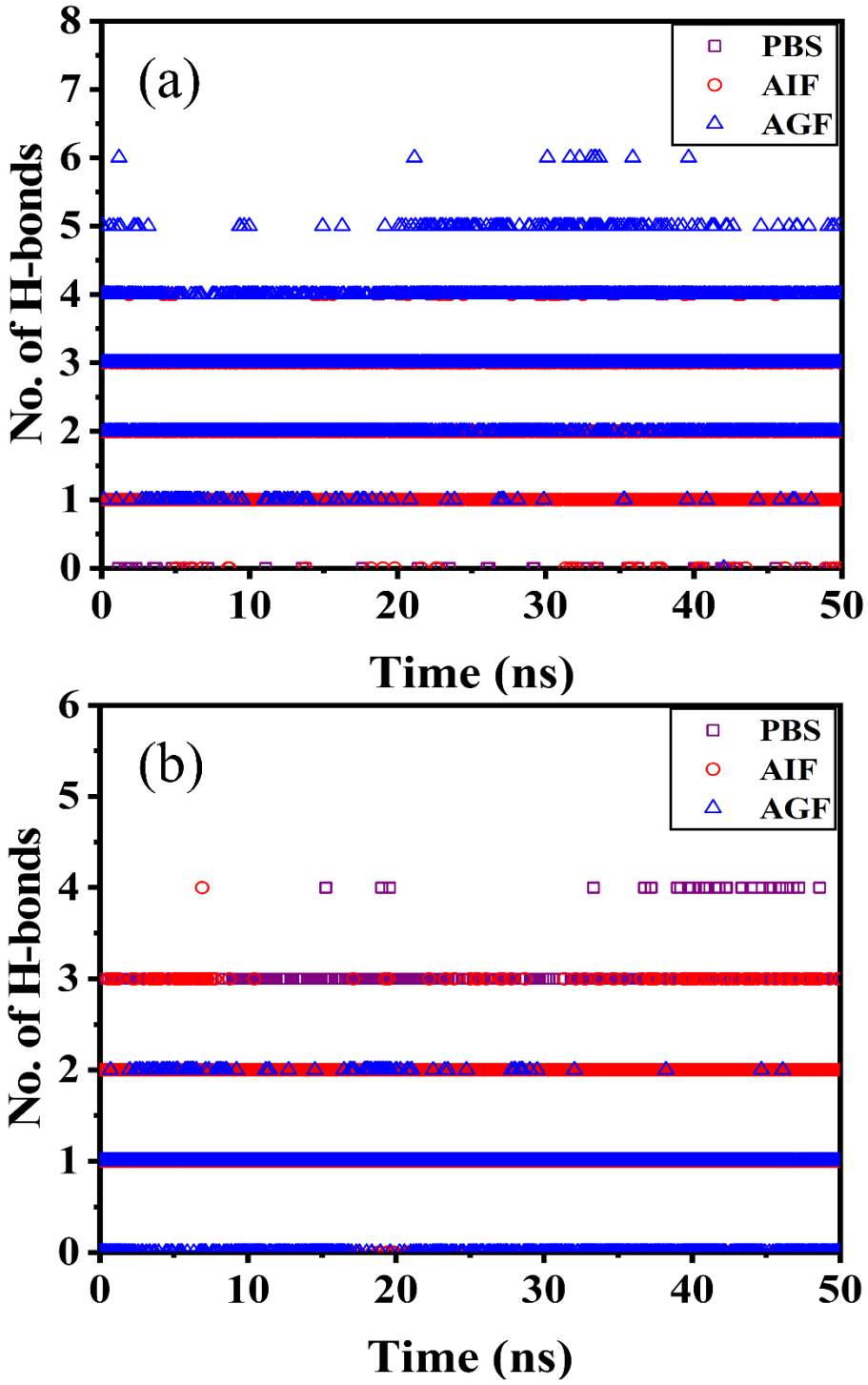


Figure S3. H-bonding between the lutein to other components system (a) lutein to OST-modified starch (b) lutein to solvents